

Context : act as data base professional, use Microsoft access to design and create a database for catering company trace rock Melon, Honeydew, Watermelon, Pineapple 4 types fruit daily net usage. Please use plain English explain to a entry level people in step by step to build the data base. the access file need able to have daily entry in daily production in net weight of 4 types fruit by kgs in Economic class and business class, daily wastage record in daily entry for 4 types fruit by kgs, can record other usage such as SPML, Charter flight, Function, Visitors by business class type, the file can record sample

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Microsoft Access Database Design Guide

Catering Fruit Usage & Yield Tracking System

This guide will help you build a **professional Microsoft Access database** step by step using very simple English.

The database will track:

- 4 fruits:
 - Watermelon
 - Honeydew
 - Rockmelon
 - Pineapple

The system will record:

- ✓ Daily production usage
- ✓ Economy & Business class usage
- ✓ Wastage
- ✓ SPML usage
- ✓ Charter / Function / Visitor usage
- ✓ Substitution usage with comments
- ✓ Fruit yield testing (Raw Weight vs Net Weight)
- ✓ Daily forecast
- ✓ Reports for:
 - Daily usage

- Weekly usage
- Monthly usage
- Yield %
- Wastage %

PART 1 — PLAN THE DATABASE

Before building Access, first understand the structure.

Think like this:

What You Need	In Access
Store data	Tables
Type daily records	Forms
Calculate totals	Queries
Print/export summary	Reports

PART 2 — DATABASE STRUCTURE

You will create these tables:

Table Name	Purpose
tblFruit	List of fruits
tblDailyProduction	Daily fruit usage
tblDailyWastage	Daily wastage
tblOtherUsage	SPML / Charter / Visitor
tblYieldTest	Yield testing
tblForecast	Daily forecast
tblSubstitution	Additional usage notes

PART 3 — CREATE THE DATABASE

Step 1 — Open Microsoft Access

1. Open Microsoft Access
2. Click **Blank Database**
3. File Name:

Fruit_Usage_Tracking.accdb



4. Click **Create**

PART 4 — CREATE MASTER TABLE

Table 1 — tblFruit

This table stores fruit names.

Create tblFruit

Go to:

Create → Table Design

Add these fields:

Field Name	Data Type
FruitID	AutoNumber
FruitName	Short Text

Set:

- FruitID = Primary Key

Save table as:

tblFruit



Add Fruit Records

Open table and enter:

FruitID	FruitName
1	Watermelon
2	Honeydew
3	Rockmelon
4	Pineapple

PART 5 — DAILY PRODUCTION TABLE

This is the MOST IMPORTANT table.

It records daily usage.

Table 2 — tblDailyProduction

Go:

Create → Table Design

Add fields:

Field Name	Type
ProductionID	AutoNumber
ProductionDate	Date/Time
FruitID	Number
EconomyKG	Number
BusinessKG	Number
SPMLKG	Number
CharterKG	Number
FunctionKG	Number

Field Name	Type
VisitorKG	Number
SubstitutionKG	Number
SubstitutionComment	Long Text

Set:

- ProductionID = Primary Key

Save as:

tblDailyProduction



PART 6 — DAILY WASTAGE TABLE

Table 3 — tblDailyWastage

Create fields:

Field Name	Type
WastageID	AutoNumber
WasteDate	Date/Time
FruitID	Number
WasteKG	Number
WasteReason	Long Text

Save as:

tblDailyWastage



PART 7 — OTHER USAGE TABLE

This can separate special usage if you want more detailed tracking.

Table 4 — tblOtherUsage

Field Name	Type
OtherUsageID	AutoNumber
UsageDate	Date/Time
FruitID	Number
UsageType	Short Text
UsageKG	Number
Comments	Long Text

Examples of UsageType:

- SPML
- Charter
- Function
- Visitor

Save as:

tblOtherUsage



PART 8 — YIELD TEST TABLE

VERY IMPORTANT.

This calculates fruit wastage percentage.

Table 5 — tblYieldTest

Field Name	Type
YieldTestID	AutoNumber
TestDate	Date/Time
FruitID	Number
RawWeightKG	Number

Field Name	Type
NetWeightKG	Number
YieldPercent	Calculated
WastePercent	Calculated
TestedBy	Short Text
Comments	Long Text

Formula for Yield %

In calculated field:

```
([NetWeightKG]/[RawWeightKG])*100
```



Formula for Waste %

```
100-([YieldPercent])
```



Save as:

```
tblYieldTest
```



PART 9 — FORECAST TABLE

Table 6 — tblForecast

Field Name	Type
ForecastID	AutoNumber
ForecastDate	Date/Time
FruitID	Number
ForecastKG	Number

Field Name	Type
Notes	Long Text
Save as:	
tblForecast	

PART 10 — SUBSTITUTION TABLE

This records additional usage adjustments.

Table 7 — tblSubstitution

Field Name	Type
SubstitutionID	AutoNumber
SubstitutionDate	Date/Time
FruitID	Number
AdditionalKG	Number
Reason	Long Text

Save as:	
tblSubstitution	

PART 11 — CREATE RELATIONSHIPS

This is VERY IMPORTANT.

Go to:

Database Tools → Relationships

Add all tables.

Connect:

From	To
tblFruit.FruitID	tblDailyProduction.FruitID
tblFruit.FruitID	tblDailyWastage.FruitID
tblFruit.FruitID	tblYieldTest.FruitID
tblFruit.FruitID	tblForecast.FruitID
tblFruit.FruitID	tblSubstitution.FruitID

Tick:



Enforce Referential Integrity

Save relationships.

PART 12 — CREATE ENTRY FORMS

Forms allow easy daily data entry.

Form 1 — Daily Production Entry Form

Go:

Create → Form Wizard

Select:

tblDailyProduction



Include all fields.

Choose:

- Columnar Form

Save as:

frmDailyProduction



Improve the Form

In Design View:

Add:

- Title
 - Buttons:
 - Save Record
 - New Record
 - Close Form
-

Add Fruit Dropdown

For FruitID:

1. Delete textbox
2. Add Combo Box
3. Use values from:

tblFruit



This gives dropdown:

- Watermelon
 - Honeydew
 - Rockmelon
 - Pineapple
-

PART 13 — CREATE YIELD TEST ENTRY FORM

Form 2 — frmYieldTest

This is important for QA and costing.

Include:

- Test Date
- Fruit
- Raw Weight
- Net Weight
- Yield %
- Waste %

Save as:

frmYieldTest



PART 14 — CREATE FORECAST ENTRY FORM

Form 3 — frmForecast

Fields:

- Forecast Date
- Fruit
- Forecast KG
- Notes

Save:

frmForecast



PART 15 — CREATE WASTAGE ENTRY FORM

Form 4 — frmWastage

Fields:

- Waste Date
- Fruit
- Waste KG
- Reason

Save:

frmWastage



PART 16 — CREATE QUERIES

Queries calculate totals automatically.

Query 1 — Daily Usage Summary

Name:

qryDailyUsage



Fields:

- Date
- Fruit
- EconomyKG
- BusinessKG
- SPMLKG
- CharterKG
- FunctionKG
- VisitorKG
- Total Usage

Create calculated field:

TotalUsage:

[EconomyKG] + [BusinessKG] + [SPMLKG] + [CharterKG] + [FunctionKG] + [VisitorKG]



Query 2 — Weekly Usage

Name:

qryWeeklyUsage



Group by:

- Week
- Fruit

Sum:

- Total Usage

Query 3 — Monthly Usage

Name:

qryMonthlyUsage



Group by:

- Month
- Fruit

Sum:

- Total Usage
-

Query 4 — Average Yield %

Name:

qryAverageYield



Group by:

- Fruit

Average:

YieldPercent



Average:

WastePercent



Query 5 — Most Recent Yield %

This shows latest yield result.

Name:

qryLatestYield



Sort:

- TestDate Descending

Show:

- First record only
-

PART 17 — CREATE REPORTS

Reports are printable/exportable.

Report 1 — Daily Usage Report

Use:

qryDailyUsage



Save:

rptDailyUsage



Report 2 — Weekly Usage Report

Use:

qryWeeklyUsage



Save:

rptWeeklyUsage



Report 3 — Monthly Usage Report

Use:

qryMonthlyUsage



Save:

rptMonthlyUsage



Report 4 — Yield Report

Use:

qryAverageYield



Show:

- Average Yield %
- Average Waste %

Save:

rptYieldSummary



PART 18 — CREATE MAIN MENU

Professional databases use a Main Menu.

Create Main Form

Create → Blank Form

Add buttons:

Button	Opens
Daily Production	frmDailyProduction
Wastage Entry	frmWastage
Yield Test	frmYieldTest
Forecast Entry	frmForecast
Daily Report	rptDailyUsage
Weekly Report	rptWeeklyUsage
Monthly Report	rptMonthlyUsage
Yield Summary	rptYieldSummary

Save as:

frmMainMenu



PART 19 — MAKE DATABASE PROFESSIONAL

Recommended Improvements

1. Add Validation

Example:

Prevent negative weights.

Field Validation Rule:

`>=0`



2. Add Auto Date

Default Value:

`Date ()`



3. Add Navigation Buttons

Use:

Design → Button

4. Add Company Logo

Insert image in Main Menu.

5. Add User Tracking

Optional fields:

- EnteredBy
- EnteredDate

PART 20 — FINAL DATABASE FLOW

Your staff workflow:

Task	Form
Enter production	frmDailyProduction
Enter wastage	frmWastage
Enter yield test	frmYieldTest
Enter forecast	frmForecast
Print reports	Reports

FINAL RESULT

Your Access database will be able to:

- ✓ Record daily fruit usage
- ✓ Track Economy & Business usage separately
- ✓ Track SPML / Charter / Visitor / Function usage
- ✓ Record substitutions with comments
- ✓ Record wastage
- ✓ Track yield testing
- ✓ Calculate wastage percentage automatically
- ✓ Record forecasts
- ✓ Generate daily / weekly / monthly reports
- ✓ Export reports to Excel/PDF
- ✓ Operate professionally for catering production control

PROFESSIONAL RECOMMENDATION

For best performance:

Use This Structure:

FRONT END

Contains:

- Forms
- Queries
- Reports

BACK END

Contains:

- Tables only

This allows:

- Multiple users
 - Better speed
 - Less corruption risk
-

Suggested Future Upgrade

Later you can add:

- Supplier tracking
- Cost per KG
- Flight numbers
- Airline customer
- Stock inventory
- Barcode labels
- Dashboard charts
- Power BI reporting
- Email automation